

AYUSH KUMAR

Bhubaneswar, India

☎ +91-7667214728

✉ ayushbarnwal062@gmail.com

🌐 www.linkedin.com/in/aayu62

🐙 github.com/Aayu62

Summary

Curious and highly self-driven 3rd-year B.Tech CSE student with a strong interest in Machine Learning, emerging technologies and hands-on engineering. I enjoy exploring new tools, building practical systems, and solving problems through experimentation. Known for an optimistic mindset and rapid learning ability, I actively pursue projects that strengthen my technical depth and real-world development skills.

Education

Kalinga Institute of Industrial Technology (KIIT)

Aug 2023 – Sept 2027

B.Tech in Computer Science and Engineering

Bhubaneswar, India

- Pursuing a comprehensive curriculum in Computer Science and Engineering.
- Collaborating with professors and peers on academic and technical projects.

I.G.N.S College, Kuchaikote

June 2020 – June 2022

Senior Secondary (Physics, Chemistry, Mathematics)

Bihar, India

Certifications

- Machine Learning — Kaggle (2025)
- Python — Kaggle (2025)
- Game AI and Reinforcement Learning — Kaggle (2025)

Projects

Connect 4 Web Game with AI | *Python, Django, JavaScript, NumPy, PyTorch*

Mar 2026

- Created **Minimax tree search** with **Alpha-Beta pruning**, cutting branching factor to $O(7^{2.5})$ and AI decision time to **<100ms** across ~2,400 nodes/turn.
- Applied **automated reasoning** via **PyTorch ValueNet** (3-layer MLP, 42-dim state vectors) for **neural-guided position evaluation** over hand-crafted heuristics.
- Optimized **move ordering** (DSA) to maximize Alpha-Beta cutoffs, reducing redundant subtree evaluations by **50%**.
- **Future:** Generalize into a **multi-game AI framework** (Chess, Checkers) via a **generic game-tree interface** reusing the same Minimax + ValueNet pipeline.

Gold Price Prediction Dashboard | *Python, Pandas, Scikit-learn, Streamlit*

Mar 2026

- Analyzed 40+ years of gold price data to identify trends and demand patterns.
- Applied K-Means and DBSCAN for segmentation and anomaly detection.
- Built SVM model achieving ~50% accuracy; deployed interactive dashboard for insights.
- Engineered features from historical price and purchase data to improve model performance.
- **Future:** Use LSTM/ARIMA and integrate live price APIs.

Sentiment Analysis Web App | *FastAPI, Scikit-learn, JavaScript*

Mar 2026

- Developed ML web app using TF-IDF + SVM achieving ~65% accuracy.
- Deployed REST API with real-time predictions and integrated responsive frontend.
- End-to-end deployment with full-stack integration and improved UI/UX.
- Handled multi-line input processing with aggregated sentiment output.
- **Future:** Add neutral class, live data integration, and visual dashboards.

DevArena — Real-Time Coding Duel Platform | *Java, PostgreSQL, Judge0 API*

Jan 2026

- Built backend for real-time 1v1 coding duels with match lifecycle and winner logic based on earliest correct submission.
- Integrated Judge0 API to execute code in 50+ languages securely.
- Designed PostgreSQL schema and scalable leaderboard system supporting 1,000+ users.
- Developed REST APIs for match creation, submissions, and user interactions.

Technical Skills

Programming Languages: Python, Java, HTML/CSS, JavaScript

Machine Learning / AI: Machine Learning, Neural Networks, Federated Learning, Deep Learning Fundamentals

Libraries / Frameworks: TensorFlow, NumPy, Pandas, Matplotlib

Tools / Technologies: Git, GitHub, Postman, REST APIs, JSON, Jupyter Notebook

Databases: PostgreSQL